In the first 3 screens the sensor temperatures are displayed. If the measured temperatures are between their low and high limits, only the temperature value is displayed, otherwise 'LOW' or 'HIGH' message is added in the end of the line. In case of sensor break, only 'SENSOR BREAK' message is displayed.

The fourth screen displays the states of the compressor motor, fan motor, the phase sequence and remote disable. This information is compiled from the digital inputs. (Digital Inputs 3 to 8)

In the first two line of the fifth screen the total and ECO operation durations of the dryer are displayed. These values can not be reset. The third line displays the run time since the last maintenance. The fourth line indicates the filter usage time. If these times exceed their set values (see section 2.3.3.), i.e. general maintenance period and filter change period, a flashing warning with a message 'MAINTENANCE' or 'REPLACE FILTER' is displayed in the fourth line of the display.

The sixth screen displays the last four events that caused the dryer to be stopped automatically.

The seventh screen displays the date and time.

The eighth screen displays the states of the digital inputs and digital outputs. The letters 'L' and 'H' stands for 'not activated' and 'activated' states respectively.

In order to start the dryer, all the temperatures except the exchanger temperature must be between their low and high limits. The low pressure line temperature can be 'HIGH'. Digital inputs 3 to 8 should not be activated. If this is not the case, instead of 'READY' message, 'DISABLED' message is displayed in the normal operation screen.

When the dryer is started, the normal operation screen is displayed as shown below..

```
      SULLAIR

DEW POINT   .   16°C
READY
```

If the drain output is activated, 'DRAIN ON' message is displayed at the end of the last line. While the fan motor is running, 'FAN MOTOR IS ON' message is displayed in the second line. In ECO mode the appearance of the display is given below.

```
      SULLAIR
ECO(SEC.)   :    234
DEW POINT        8°C
RUNNING
```

The value at the end of the second row indicates the time in seconds since beginning of the ECO mode.

If the dryer is stopped automatically because of an anomaly, or manually by using button, the normal operating screen is displayed as shown below.

```
      SULLAIR

DEW POINT       16°C
DISABLED         127
```

The number at the end of the last line indicates the remaining time in seconds from the restart delay. If this number becomes zero and there is no anomaly, the dryer can be restarted.

button is used to change the temperature unit from °C to °F or vice versa. When this button is pressed, the control operation stops about 3 seconds and all the temperature values are converted to the selected unit.

The first line in the normal operation display (SULLAIR) is user configurable.

```
INLET AIR TEMP.
      20°C
EXCHANGER.TEMP.
      18°C      HIGH
```

```
TOTAL TIME  :    22.2
ECO TIME    :     8.1
RUN TIME    :    22.2
FILTER TIME:     22.2
```

```
    LOW PRES.LINE
        TEMP.
        18°C
HIGH PRES.LINE TEMP.
```

```
1. REMOTE DISABLE
2. INLET AIR TEMP.HI
3. EXCH TEMP.LO
4. COMPRESSOR FAULT
```

```
AMBIENT TEMPERATURE
      19°C
AUXILIARY TEMP.
      18°
```

```
DATE :       22.03.10
TIME :       12.30.45
DAY  :       MONDAY
```

```
COMPRESSOR OK
FAN MOTOR OK
PHASE SEQUENCE OK
REMOTE CONTROL OK
```

```
D.INPUT1:    LLLLLLLL
D.INPUT2:    LLLLLLLL
D.OUTPUT:    LLLHHLLL
```

T.No	Name	Function
13	D. OUTPUT 8	Alarm Output (Horn)
14	-	
15	COMM. For D. INPUTS	
16	COMM. For D. INPUTS	
17	COMM. For D. INPUTS	
18	D. INPUT 1	Remote Start (Press and Release)
19	D. INPUT 2	Remote Stop (Press and Release)
20	D. INPUT 3	Compressor Fault
21	D. INPUT 4	Compressor Overload
22	D. INPUT 5	Fan Fault
23	D. INPUT 6	Fan Overload
24	D. INPUT 7	Phase Sequence Error
25	D. INPUT 8	Remote Disable
26	D. INPUT 9	Fan Motor is On
27	D. INPUT 10	Configuration Enable
28	D. INPUT 11	Spare
29	D. INPUT 12	Spare
30	D. INPUT 13	Spare
31	D. INPUT 14	Spare
32	D. INPUT 15	Spare
33	D. INPUT 16	Spare

The contact rating of each output relay is 10A at 250 V AC.

Digital inputs are activated by 24V DC or AC.

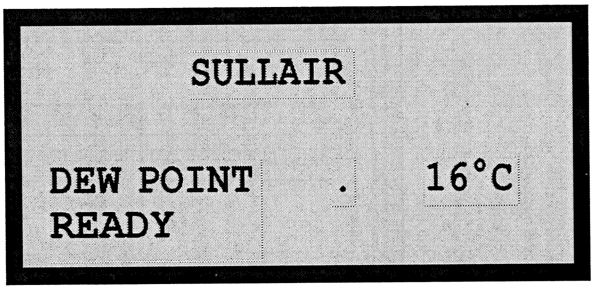
TERMINALS IN THE LOWER ROW

The first 3 terminals are for operating voltage. (Phase, Neutral and Ground) The ground terminal should be connected to the chassis of the dryer.

The last 3 terminals are for RS-485 communication line.

6.2.2 NORMAL OPERATION

When the controller is powered on it displays the type and version message and then normal operation screen is displayed as shown below.



The exchanger temperature and operation state of the dryer is displayed in this screen. Sequentially pressing button display other info related to the dryer in 8 different screens. While one screen is displayed, pressing button reverts to the normal operation screen. The views of the screens in sequence of the appearance are shown below.

The dryer is automatically stopped if an anomaly is detected. In that case, the alarm output and the alarm indicator LED on the front panel become activated. In order to restart the dryer, alarm should be acknowledged and “restart delay” period should be timed out. Pressing button acknowledges the alarm and de-energizes the alarm output and alarm LED.

and buttons are used for starting and stopping the dryer respectively. If the dryer is stopped manually, it can not be started before “restart delay” period is timed out.

6.2.1 EXTERNAL CONNECTIONS

The back panel view and the connection terminals of E-680 controller are given in *Figure 6-2*.

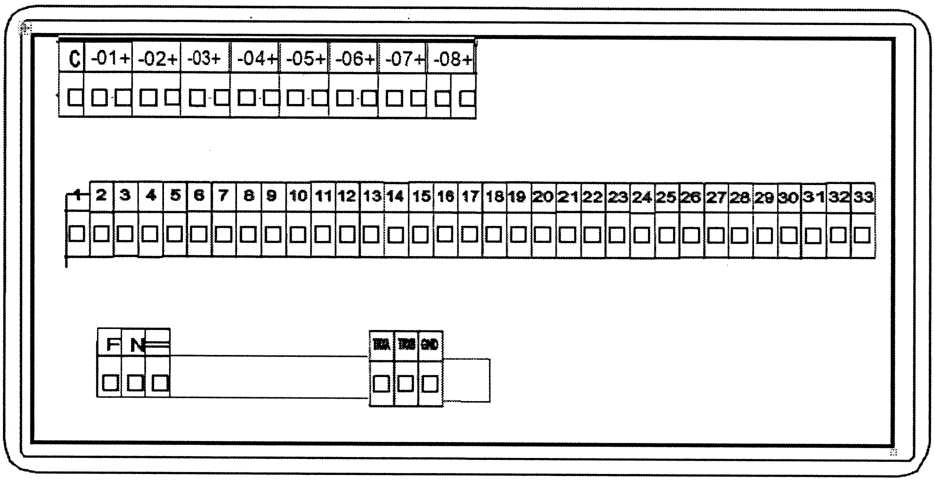


Figure 6-2: The Backpanel View of E-680 Controller

TERMINALS IN THE UPPER ROW

These terminals are used for temperature sensors (Pt-100).

1 Inlet Air Temperature.

2 Exchanger Temperature.

3 Low Pressure Line Temperature.

4 High Pressure Line Temperature.

5 Ambient Temperature.

6 Auxiliary Channel Temperature. This temperature can be monitored, but sensor break has no effect on the operation of the controller.

7 Spare Channel 1.

8 Spare Channel 2.

Spare channels are measured by the controller, but can not be monitored and sensor break for these channels has no effect on the operation of the controller.

TERMINALS IN THE MIDDLE ROW

These terminals are used for digital outputs and digital inputs. The names and the functions in the order of the terminal numbers are given in the following table.

T.No	Name	Function
1	COMM. For D. Out 1-4	
2	COMM. For D. Out 1-4	
3	D. OUTPUT 1	Compressor Motor
4	D. OUTPUT 2	Drain Output (Normal)
5	D. OUTPUT 3	Dryer is Running.
6	D. OUTPUT 4	Dryer is Stopped. In case of fault, the output flashes.
7	-	
8	COMM. For D. Out 5-8	
9	COMM. For D. Out 5-8	
10	D. OUTPUT 5	Drain Output (Inverted)
11	D. OUTPUT 6	Spare
12	D. OUTPUT 7	Spare

DIGITAL CONTROLLER E-680

(RF-A-3000 to RF-A-7300)

E-680 is designed as a controller for refrigerant type compressed air dryers. The controller has 8 temperature sensor inputs. Each channel can be configured for type J, type K thermocouple or Pt-100 resistance thermometer. These inputs are used to measure the temperatures at various points in the dryer. The controller has also 8 digital (relay) outputs and 16 digital inputs. The digital outputs are taken through the normally open contacts of the output relays. The contact rating of the output relays are 10A at 250 V AC Digital inputs are activated by 24V DC or AC.

The controller has an RS-485 communication interface that can be used for remotely monitoring channel temperatures, set points, input and output states. Modbus RTU protocol is used for communication.

The front panel of the controller contains a four line 20 character LCD display and 10 buttons that are used in configuration and manual control operations.

The dimensions of the controller are 3.8 x 7.6 in / 96 x 192 mm (front panel) with a depth of 4.3 in / 110 mm. The panel cutout should be 3.5 x 7.2 in / 90x185 mm. The operating voltage of the controller is 20 - 60V AC or 20 - 85V DC.

6.2 Operation for E-680

The front panel view of E-680 controller is given in Figure 6-1. The front panel of the controller contains a four line 20 character LCD display, 9 buttons and one alarm indicator LED.

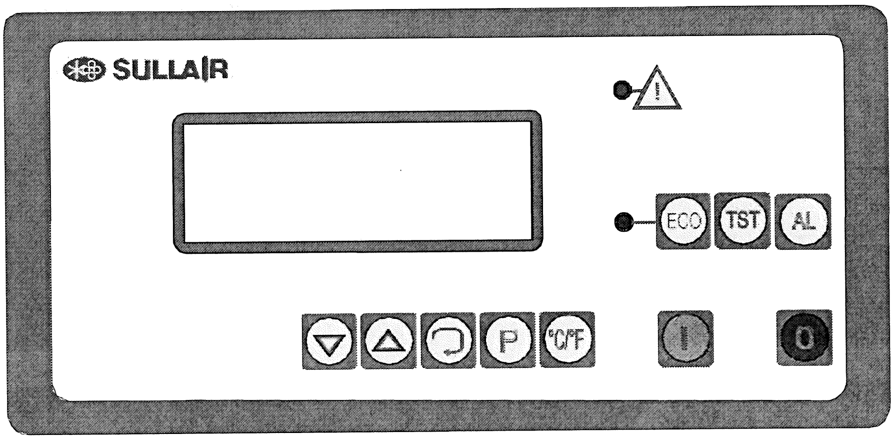


Figure 6-1: The Front Panel View of E-680 Controller

The buttons below the LCD display (, , , and) are used in configuration operations.

button is used for manual control of the drain output. While in normal operation, the drain output is controlled according to the configured “drain on” and “drain off” periods. When button is pressed the drain output is activated even if the dryer is in off state.

Energy Saving Mode

Shows if the drain system is activated

°C **Celcius Unit Mode**

Indicates that Celsius temperature unit is selected.

°F **Fahrenheit Unit Mode**

Indicates that Fahrenheit temperature unit is selected.

Compressor Standby Mode
This mode shows that the dryer is ready for drying operation.

Service Standby Mode

This mode shows that the dryer
Is ready for drying operation.

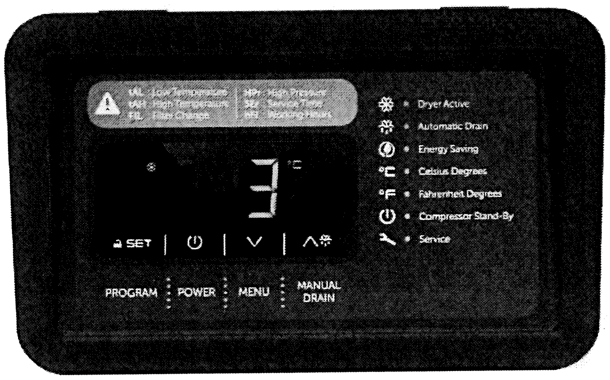
Alarm Code	Alarm Description	Reason for Alarm
tAL	Low Temperature Alarm	Refrigerant line temperature is lower than specified set values
tAH	High Temperature Alarm	Refrigerant line temperature is higher than specified set values
FIL	Filter Change Alarm	Filter element needs to be replaced
SEr	General Service Alarm	General service time of the dryer
HPr	High Pressure Alarm	Refrigerant high line pressure is higher than specified set values
Pr1	Temperature Probe Alarm	Temperature sensor is faulty.
hFI	Working Hours Alarm	The dryer working hours allowed has been reached.

CONTROLLERS

6.1 DIGITAL CONTROLLER (Digi-Pro)

(RF-A-10 to RF-A-2300)

With the Digi-Pro series controllers, air dryers have outstanding technology for both functionality and dynamism, as well as appearance. The multi-functional display provides an accurate digital dew point display as well as coded alarm monitoring of the refrigerant dryer.



Digital Controller (Digi-Pro)

Digital controller advantages;

- Digital dew point monitoring
- Energy-saving mode display
- Periodic maintenance interval display
- Status report
- Run time meter
- Fahrenheit and Centigrade selection

3.1.1 OPERATION

Using the Digi-Pro controller as shown in the picture here is the menu below;

SET

Program

To modify the parameter, press and release button set. The menu is used by service team.

Power

This button is used for starting and stopping the dryer. Press and hold for 4 seconds to start or stop.

Menu

These buttons are used to navigate between screens and adjust values.

Manual Drain

This button is used for manual control of the drain output. Press and hold for 4 seconds to drain manually.

3.1.2 MODE DISPLAY

Dryer Active Mode

This mark indicates that the dryer is performed in active state and drying.

Automatic Drain Mode

Shows if the drain system is activated.

5.3 STARTUP AND SHUT DOWN PROCEDURE

⚠ WARNING

For short periods of inactivity, (max 2-3 days) we recommend that power is maintained to the dryer and the control panel. Otherwise, before re-starting the dryer, it is necessary to wait at least 2-3 hours for the compressor crankcase heater to heat the oil of the compressor.

SEC 5.3.1 START-UP:

1. Check the condenser for cleanliness (Air-Cooled).
2. Ensure the cooling water flow and temperature is adequate (Water-Cooled).
3. Check that the mains Disconnect ON/OFF switch is ON.
4. Switch ON the dryer pressing the button "I - ON" of the ON/OFF switch
5. Ensure that electronic Monitor is ON.
6. Wait a few minutes; verify that the Dew Point temperature displayed on electronic instru- ment is dropping and condensate drain is discharging .

SEC 5.3.2 SHUT DOWN

1. Shut down the air compressor.
2. After few minutes, shut down the dryer pressing the button "0 - OFF" of the digital controller

NOTE

A Dew Point within 32° F and 50°F/0°C and +10°C displayed on Air Dryer Controller is correct according to the possible working conditions (flow-rate, temperature of the incoming air, ambient temperature, etc.).

During the operation, the refrigeration compressor will cycle ON/OFF on RF-A series. The dryer must remain ON during the full usage period of the compressed air, even if the air compressor works

⚠ WARNING

The number of start/stop must be no more than 6 time per hour. The dryer must stop running for at least 5 minutes before being started up again.

The user is responsible for compliance with these rules. Frequent start/Stop may cause irreparable damage to refrigeration compressor.

OPERATION

5.1 STEPS TO UNDERTAKE BEFORE OPERATING

1. Read this manual completely.
2. Review all safety precautions.
3. Use recommended pipe sizes as per specifications.
4. DO NOT operate the dryer at pressures above the maximum specified on the dryer label (check the technical specifications).
5. DO NOT operate the dryer in temperatures above 120°F / 50°C degrees
6. DO NOT operate the dryer with inlet air temperatures above 150°F / 65°C.

5.2 OPERATING PROCEDURES

This procedure should be followed on first start-up, after periods of extended shutdown or following maintenance procedures. Qualified personnel must perform the start-up.

SEQUENCE OF OPERATIONS

1. Ensure that all the steps of the "Installation" chapter have been observed.
2. Ensure that the connection to the compressed air system is correct and that the piping is suitably fixed and supported.
3. Ensure that the condensate drain pipe is properly fastened and connected to a collection system.
4. Ensure that the by-pass system (if installed) is closed and the dryer is isolated.
5. Ensure that the manual valve of the condensate drain circuit is open.
6. Remove any packaging and other material which could obstruct the area around the dryer.
7. Activate the mains switch.
8. Turn on the main switch - pos. 1 on the control panel.
9. Check that the mains power light of the ON/OFF button is ON.
10. Wait at least two hours before starting the dryer (compressor crankcase heater must heat the oil of the compressor)
11. Ensure the cooling water flow and temperature is adequate (Water-Cooled).
12. Switch ON the dryer pressing the button "I - ON" of the ON/OFF switch on the digital controller
13. Ensure the data plate flow rate match with plant consumption.
14. If the dryer does not start to run check the monitor on the Microprocessor. If the Phase error is mentioned on the screen change the Phases and re-start the dryer.
15. Allow the dryer temperature to stabilize

    10- 15 minutes
16. Slowly open the air inlet valve.
17. Slowly open the air outlet valve.
18. Slowly close the central by-pass valve of the system (if optional By-pass valves are installed).
19. Check the piping for air leakage.
20. Ensure the drain is regularly cycling - wait for its first interventions.

The installer shall obtain all permits and inspections required for the work, and shall pay all costs and fees thereof.

4.6 OPERATING PRECAUTIONS

WARNING

Verify that the operating parameters match with the nominal values stated on the data nameplate of the dryer (voltage, frequency, air pressure, air temperature, ambient temperature, etc.).

This dryer has been thoroughly tested, packaged and inspected prior to shipment. Nevertheless, the unit could be damaged during transportation, check the integrity of the dryer during first start-up and monitor operation during the first hours of operation. Qualified personnel must perform the first start-up. When installing and operating this equipment, comply with all National Electrical Code and any applicable federal, state and local codes. Who is operating the unit is responsible for the proper and safe operation of the dryer. Never operate equipment with panels removed.

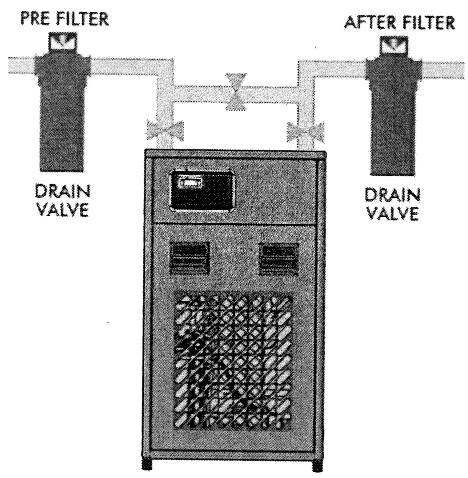
4.7 INSTALLATION

In addition to the general mechanical construction procedures and local regulations, the following instructions need to be emphasized:

1. Only authorized, trained and skilled engineers should install the compressed air dryer.
2. Safety devices, protecting covers or insulations in the dryer are never to be removed or modified. Each pressure vessel or accessory installed outside the dryer with compressed air (Any pressure above atmospheric pres- sure) must be fitted with individual pressure relief safety valves.
3. Install one (1) air by-pass valve and two (2) switch off valves in the line to isolate the dryer to allow easy maintenance and for possible isolation of the dryer without interrupt- ing the compressed air flow.

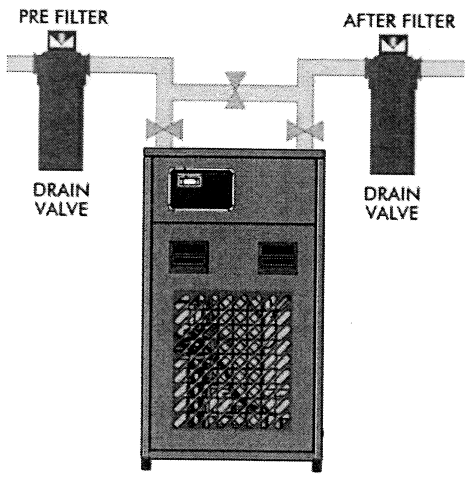
In Line Filters: install one compressed air filter in line (particulate filter) before the dryer to protect it against dirt and possible clogging of heat exchanger. Install an After-filter (coalescing filter) that can act as a backup during short periods when the by-pass valve is open to perform maintenance on the dryer. Contact your Sullair representative/dealer for suitable filters.

5. Water Filter: For dryers with water cooled condensers, a water filter should be installed in the supply water piping. (Water Cooled Units Only)



DIGITAL NON-CYCLING
REFRIGERATED AIR DRYER

6. The interconnecting piping must be of the correct size, pressure rating and match that of the dryer Inlet and Outlet pipe sizes. DO NOT use pipes of lesser diameter as this will lead to additional pressure loss and will affect the performance of your air dryer. If you are in doubt, contact your Sullair distributor. It is recommended to use the same pipe size as that of the dryer Inlet and Outlet.



DIGITAL NON-CYCLING
REFRIGERATED AIR DRYER

INSTALLATION

4.1 RECEVING AND INSPECTION

Sullair dryers are factory tested prior to shipment. However, during shipment, there are chances that the dryer was mishandled or certain parts were broken or might have come loose. In order to ensure that the dryer you have received is fit for installation and operation, we recommend you take a few minutes to inspect the dryer for any physical abuse or damage during shipment.

If you notice any damage to the shipment, or if the blue beads in the Tip-n-Tell label are in the arrow (read instructions on the label attached to the box) we recommend you take the

following steps to ensure proper arrangements can be made to address the issue.

1. Immediately file a complaint with the shipping company.
2. Contact Sullair or Sullair distributor and inform them of the damage. (Email or fax the copy of the shipper claim and acknowledgement)

If there is no physical damage to the dryer –

1. Remove all the crating and packaging.
2. Inspect the dryer for electrical and piping connection and make sure that they are all tight.
3. Check the data label, your packing slip and make sure that it is the correct unit you had ordered.
4. Follow the recommended safety procedures and get the dryer ready for installation.

4.2 TRANSPORTATION

1. Use care and caution when transporting the dryer. Avoid sudden jerks, tilting, dropping and other physical abuse.
2. A forklift can be used to transport the dryer provided the forks are long enough to support the full width or length. Caution must be used throughout the move.

4.3 INSTALLATION LAYOUT REQUIREMENTS

1. The dryer must be installed horizontally. A minimum of at least 1.5 feet / 0.5 meters clearance around the dryer is necessary to allow free air circulation and easy access for servicing.
2. The ambient temperature in the room should not exceed 120°F / 49°C and should not be below 40°F / 4°C, taking into account the heat radiated by the dryer. (About 18 watts for each SCFM under ISO 7183-5 condition or 40 watt for each liter/sec under ISO 7183- A condition).

4.4 PIPE/CONNECTION REQUIREMENTS

Refer to Product Models and Specifications—*Table 2-2*, *Table 2-3*, *Table 2-4*, *Table 2-5*.

4.5 ELECTRICAL CONNECTION REQUIREMENTS

The owner shall furnish all labor, materials, equipment and services necessary for and reasonably incidental to complete the installation of all electrical as shown on the drawings.The electrical installation and connection shall be made in strict accordance with the requirements of any and all City, County, State or Federal codes of Law having jurisdiction, the requirements and recommendations of the Board of Fire Underwriters, including all amendments and / or additions to the said codes, laws, requirements, and recommendations, the requirements and recommendations of the local utility, the Owner, and the Standard Building Code.

AIR COOLED MODEL NUMBER	SULLAIR PART NUMBER	Width		Depth		Height		Weight	
		in	mm	in	mm	in	mm	lbs	kg
RF-A-800-400-3-50-AC-NPT	02250225-588	31.44	798	37.32	948	57.45	1459,5	595	270
RF-A-1000-400-3-50-AC-NPT	02250225-589	31.44	798	37.32	948	57.45	1459,5	629	285
RF-A-1460-400-3-50-AC-NPT	02250225-590	30.63	778	45.79	1163	67.81	1722	864	392
RF-A-1600-400-3-50-AC-NPT	02250225-591	30.63	778	45.79	1163	67.81	1722	904	410
RF-A-2000-400-3-50-AC-FLANGE	02250225-592	33.35	847	55	1397	69.65	1769	1085	492
RF-A-2300-400-3-50-AC-FLANGE	02250225-593	33.35	847	55	1397	69.65	1769	1146	520
RF-A-3000-400-3-50-AC-FLANGE	02250225-594	42.40	1077	57.76	1467	75.83	1926	1535	696
RF-A-3460-400-3-50-AC-FLANGE	02250225-595	42.40	1077	57.76	1467	75.83	1926	1583	718
RF-A-4000-400-3-50-AC-FLANGE	02250225-596	41.81	1062	86.13	2188	75.75	1924	1984	900
RF-A-4600-400-3-50-AC-FLANGE	02250225-597	41.81	1062	86.13	2188	75.75	1924	2035	923
RF-A-5200-400-3-50-AC-FLANGE	02250225-598	47.24	1200	88.46	2247	80.46	2044	2150	975
RF-A-6100-400-3-50-AC-FLANGE	02250225-599	47.24	1200	88.46	2247	80.46	2044	2450	1111
RF-A-7300-400-3-50-AC-FLANGE	02250225-600	61.02	1550	104.33	2650	82.68	2100	3090	1402

Table 2-4: Models 60HZ

AIR COOLED MODEL NUMBER	SULLAIR PART NUMBER	Width		Depth		Height		Weight	
		in	mm	in	mm	in	mm	lbs	kg
RF-A-4600-460-3-60-AC-FLANGE	02250225-571	41.81	1062	86.13	2188	75.75	1924	2035	923
RF-A-5200-460-3-60-AC-FLANGE	02250225-572	47.24	1200	88.46	2247	80.46	2044	2150	975
RF-A-6100-460-3-60-AC-FLANGE	02250225-573	47.24	1200	88.46	2247	80.46	2044	2450	1111
RF-A-7300-460-3-60-AC-FLANGE	02250225-574	61.02	1550	104.33	2650	82.68	2100	3090	1402

Table 2-5: Models 50Hz

AIR COOLED MODEL REF	SULLAIR PART NUMBER	Width		Depth		Height		Weight	
		in	mm	in	mm	in	mm	lbs	kg
RF-A-10-230-1-50-AC-NPT	02250225-575	13.91	353	16.65	423	21.93	557	72	33
RF-A-20-230-1-50-AC-NPT	02250225-576	13.91	353	16.65	423	21.93	557	72	33
RF-A-30-230-1-50-AC-NPT	02250225-577	13.91	353	16.65	423	21.93	557	72	33
RF-A-50-230-1-50-AC-NPT	02250225-578	17.85	453	18.64	473	32.75	832	113	51
RF-A-75-230-1-50-AC-NPT	02250225-579	17.85	453	18.64	473	32.75	832	117	53
RF-A-100-230-1-50-AC-NPT	02250225-580	17.85	453	18.64	473	32.75	832	122	55
RF-A-125-230-1-50-AC-NPT	02250225-581	19.84	504	21.81	554	34.40	874	172	78
RF-A-175-230-1-50-AC-NPT	02250225-582	19.84	504	21.81	554	34.40	874	183	83
RF-A-210-230-1-50-AC-NPT	02250225-583	19.84	504	21.81	554	34.40	874	190	86
RF-A-275-230-1-50-AC-NPT	02250225-584	25.51	648	26.69	678	45.56	1157.5	354	161
RF-A-350-230-1-50-AC-NPT	02250225-585	25.51	648	26.69	678	45.56	1157.5	364	165
RF-A-500-230-1-50-AC-NPT	02250225-586	28.66	728	37.32	948	53.91	1369,5	485	220
RF-A-650-230-1-50-AC-NPT	02250225-587	28.66	728	37.32	948	53.91	1369,5	507	230

Table 2-4: Models 60HZ

AIR COOLED MODEL NUMBER	SULLAIR PART NUMBER	Width		Depth		Height		Weight	
		in	mm	in	mm	in	mm	lbs	kg
RF-A-10-230-1-60-AC-NPT	02250225-548	13.91	353	16.65	423	21.93	557	72	33
RF-A-20-230-1-60-AC-NPT	02250225-549	13.91	353	16.65	423	21.93	557	72	33
RF-A-30-230-1-60-AC-NPT	02250225-551	13.91	353	16.65	423	21.93	557	72	33
RF-A-50-230-1-60-AC-NPT	02250225-552	17.85	453	18.64	473	32.75	832	113	51
RF-A-75-230-1-60-AC-NPT	02250225-553	17.85	453	18.64	473	32.75	832	117	53
RF-A-100-230-1-60-AC-NPT	02250225-554	17.85	453	18.64	473	32.75	832	122	55
RF-A-125-230-1-60-AC-NPT	02250225-555	19.84	504	21.81	554	34.40	874	172	78
RF-A-175-230-1-60-AC-NPT	02250225-556	19.84	504	21.81	554	34.40	874	183	83
RF-A-210-230-1-60-AC-NPT	02250225-557	19.84	504	21.81	554	34.40	874	190	86
RF-A-275-230-1-60-AC-NPT	02250225-558	25.51	648	26.69	678	45.56	1157.5	354	161
RF-A-350-230-1-60-AC-NPT	02250225-559	25.51	648	26.69	678	45.56	1157.5	364	165
RF-A-500-230-1-60-AC-NPT	02250225-560	28.66	728	37.32	948	53.91	1369,5	485	220
RF-A-650-230-1-60-AC-NPT	02250225-561	28.66	728	37.32	948	53.91	1369,5	507	230
RF-A-800-460-3-60-AC-NPT	02250225-562	31.44	798	37.32	948	57.45	1459,5	595	270
RF-A-1000-460-3-60-AC-NPT	02250225-563	31.44	798	37.32	948	57.45	1459,5	629	285
RF-A-1460-460-3-60-AC-NPT	02250225-564	30.63	778	45.79	1163	67.81	1722	864	392
RF-A-1600-460-3-60-AC-NPT	02250225-565	30.63	778	45.79	1163	67.81	1722	904	410
RF-A-2000-460-3-60-AC-FLANGE	02250225-566	33.35	847	55	1397	69.65	1769	1085	492
RF-A-2300-460-3-60-AC-FLANGE	02250225-567	33.35	847	55	1397	69.65	1769	1146	520
RF-A-3000-460-3-60-AC-FLANGE	02250225-568	42.40	1077	57.76	1467	75.83	1926	1535	696
RF-A-3460-460-3-60-AC-FLANGE	02250225-569	42.40	1077	57.76	1467	75.83	1926	1583	718
RF-A-4000-460-3-60-AC-FLANGE	02250225-570	41.81	1062	86.13	2188	75.75	1924	1984	900

Table 2-3: DP Power 50H

GROUP	STANDARD VOLTAGE	Capacity SCFM @ Rated Conditions	AIR COOLED MODEL NUMBER	SULLAIR PART NUMBER	Rated Flow	Pressure Drop	Absorbed Power	Port Size
					(scfm)	(psig)	(kW)	(in)
Non- Cycling	230/1/50	13	RF-A-10-230-1-50-AC-NPT	02250225-575	13	<3	0,32	1/2" NPT
Non- Cycling	230/1/50	22	RF-A-20-230-1-50-AC-NPT	02250225-576	22	<3	0,33	1/2" NPT
Non- Cycling	230/1/50	31	RF-A-30-230-1-50-AC-NPT	02250225-577	31	<3	0,36	1/2" NPT
Non- Cycling	230/1/50	59	RF-A-50-230-1-50-AC-NPT	02250225-578	59	<3	0,37	3/4" NPT
Non- Cycling	230/1/50	91	RF-A-75-230-1-50-AC-NPT	02250225-579	91	<3	0,59	3/4" NPT
Non- Cycling	230/1/50	112	RF-A-100-230-1-50-AC-NPT	02250225-580	112	<3	0,68	3/4" NPT
Non-Cycling	230/1/50	124	RF-A-125-230-1-50-AC-NPT	02250225-581	124	<3	0,82	1 1/2" NPT
Non- Cycling	230/1/50	179	RF-A-175-230-1-50-AC-NPT	02250225-582	179	<3	1,07	1 1/2" NPT
Non- Cycling	230/1/50	221	RF-A-210-230-1-50-AC-NPT	02250225-583	221	<3	1,19	1 1/2" NPT
Non- Cycling	230/1/50	291	RF-A-275-230-1-50-AC-NPT	02250225-584	291	<3	1,23	2" NPT
Non- Cycling	230/1/50	366	RF-A-350-230-1-50-AC-NPT	02250225-585	366	<3	1,32	2" NPT
Non- Cycling	230/1/50	547	RF-A-500-230-1-50-AC-NPT	02250225-586	547	<3	1,84	2" NPT
Non- Cycling	230/1/50	706	RF-A-650-230-1-50-AC-NPT	02250225-587	706	<3	2,35	2" NPT
Non- Cycling	400/3/50	816	RF-A-800-400-3-50-AC-NPT	02250225-588	816	<3	2,97	3" NPT
Non- Cycling	400/3/50	1059	RF-A-1000-400-3-50-AC-NPT	02250225-589	1059	<3	3,38	3" NPT
Non- Cycling	400/3/50	1471	RF-A-1460-400-3-50-AC-NPT	02250225-590	1471	<3	4,85	3" NPT
Non- Cycling	400/3/50	1632	RF-A-1600-400-3-50-AC-NPT	02250225-591	1632	<3	4,86	3" NPT
Non- Cycling	400/3/50	1959	RF-A-2000-400-3-50-AC-FLANGE	02250225-592	1959	<3	5,61	DN100
Non- Cycling	400/3/50	2303	RF-A-2300-400-3-50-AC-FLANGE	02250225-593	2303	<3	6,33	DN100
Non- Cycling	400/3/50	2991	RF-A-3000-400-3-50-AC-FLANGE	02250225-594	2991	<3	7,92	DN100
Non- Cycling	400/3/50	3441	RF-A-3460-400-3-50-AC-FLANGE	02250225-595	3441	<3	9,22	DN100
Non- Cycling	400/3/50	4103	RF-A-4000-400-3-50-AC-FLANGE	02250225-596	4103	<3	11,66	DN150
Non- Cycling	400/3/50	4632	RF-A-4600-400-3-50-AC-FLANGE	02250225-597	4632	<3	12,27	DN150
Non- Cycling	400/3/50	5294	RF-A-5200-400-3-50-AC-FLANGE	02250225-598	5294	<3	14,71	DN150
Non- Cycling	400/3/50	6176	RF-A-6100-400-3-50-AC-FLANGE	02250225-599	6176	<3	15,32	DN200
Non- Cycling	400/3/50	7353	RF-A-7300-400-3-50-AC-FLANGE	02250225-600	7353	<3	18,57	DN200

Table 2-2: DP Power 60H

GROUP	STANDARD VOLTAGE	Capacity SCFM @ Rated Conditions	AIR COOLED MODEL NUMBER	SULLAIR PART NUMBER	Rated Flow	Pressure Drop	Absorbed Power	Port Size
					(scfm)	(psig)	(kW)	(in)
Non- Cycling	230/1/60	13	RF-A-10-230-1-60-AC-NPT	02250225-548	13	<3	0,37	1/2" NPT
Non- Cycling	230/1/60	22	RF-A-20-230-1-60-AC-NPT	02250225-549	22	<3	0,38	1/2" NPT
Non- Cycling	230/1/60	31	RF-A-30-230-1-60-AC-NPT	02250225-551	31	<3	0,44	1/2" NPT
Non- Cycling	230/1/60	59	RF-A-50-230-1-60-AC-NPT	02250225-552	59	<3	0,45	3/4" NPT
Non- Cycling	230/1/60	91	RF-A-75-230-1-60-AC-NPT	02250225-553	91	<3	0,67	3/4" NPT
Non- Cycling	230/1/60	112	RF-A-100-230-1-60-AC-NPT	02250225-554	112	<3	0,77	3/4" NPT
Non-Cycling	230/1/60	124	RF-A-125-230-1-60-AC-NPT	02250225-555	124	<3	1,00	1 1/2" NPT
Non- Cycling	230/1/60	179	RF-A-175-230-1-60-AC-NPT	02250225-556	179	<3	1,28	1 1/2" NPT
Non- Cycling	230/1/60	221	RF-A-210-230-1-60-AC-NPT	02250225-557	221	<3	1,48	1 1/2" NPT
Non- Cycling	230/1/60	291	RF-A-275-230-1-60-AC-NPT	02250225-558	291	<3	1,53	2" NPT
Non- Cycling	230/1/60	366	RF-A-350-230-1-60-AC-NPT	02250225-559	366	<3	2,05	2" NPT
Non- Cycling	230/1/60	547	RF-A-500-230-1-60-AC-NPT	02250225-560	547	<3	2,30	2" NPT
Non- Cycling	230/1/60	706	RF-A-650-230-1-60-AC-NPT	02250225-561	706	<3	3,03	2" NPT
Non- Cycling	460/3/60	816	RF-A-800-460-3-60-AC-NPT	02250225-562	816	<3	3,64	3" NPT
Non- Cycling	460/3/60	1059	RF-A-1000-460-3-60-AC-NPT	02250225-563	1059	<3	4,09	3" NPT
Non- Cycling	460/3/60	1471	RF-A-1460-460-3-60-AC-NPT	02250225-564	1471	<3	6,14	3" NPT
Non- Cycling	460/3/60	1632	RF-A-1600-460-3-60-AC-NPT	02250225-565	1632	<3	6,17	3" NPT
Non- Cycling	460/3/60	1959	RF-A-2000-460-3-60-AC-FLANGE	02250225-566	1959	<3	6,77	DN100
Non- Cycling	460/3/60	2303	RF-A-2300-460-3-60-AC-FLANGE	02250225-567	2303	<3	8,44	DN100
Non- Cycling	460/3/60	2991	RF-A-3000-460-3-60-AC-FLANGE	02250225-568	2991	<3	10,83	DN100
Non- Cycling	460/3/60	3441	RF-A-3460-460-3-60-AC-FLANGE	02250225-569	3441	<3	12,37	DN100
Non- Cycling	460/3/60	4103	RF-A-4000-460-3-60-AC-FLANGE	02250225-570	4103	<3	15,37	DN150
Non- Cycling	460/3/60	4632	RF-A-4600-460-3-60-AC-FLANGE	02250225-571	4632	<3	16,62	DN150
Non- Cycling	460/3/60	5294	RF-A-5200-460-3-60-AC-FLANGE	02250225-572	5294	<3	19,84	DN150
Non- Cycling	460/3/60	6176	RF-A-6100-460-3-60-AC-FLANGE	02250225-573	6176	<3	19,94	DN200
Non- Cycling	460/3/60	7353	RF-A-7300-460-3-60-AC-FLANGE	02250225-574	7353	<3	26,17	DN200

SPECIFICATIONS

3.1 SPECIFICATIONS

Pay attention to the Minimum and Maximum operating conditions before installing and operating the dryer.
RF-A dryer capacities are designed in accordance to CAGI standards. Variances from standards will directly affect performance.

If your application does not match the above criteria, contact your Sullair distributor and they will be able to provide the right dryer for your application.

Normal Operating Pressure	100 PSIG / 6.89 Bar
Normal Operating Temperature	95°F / 35°C
Normal Ambient Temperature	77°F / 25°C
Maximum Operating Pressure	230 PSIG / 16 Bar
Minimum Operating Pressure	58 PSIG / 4 Bar
Maximum Ambient Temperature	113°F / 45°C
Minimum Ambient Temperature	40°F / 4°C
Maximum Operating Temperature	122°F / 50°C

The Microprocessor allows the dryer to save energy when there is no flow in the dryer. It is possible to monitor major dryer failures on the Microprocessor. It allows the operators to monitor the following:

- Evaporation temperature
- Inlet air temperature
- Ambient temperature
- Refrigerant gas high and low temperatures
- Fan is working properly
- Compressor is working properly
- Condenser is blocked
- Power Phases are correctly connected
- Drain function
- Total working hours
- Total economy hours
- Real date and time

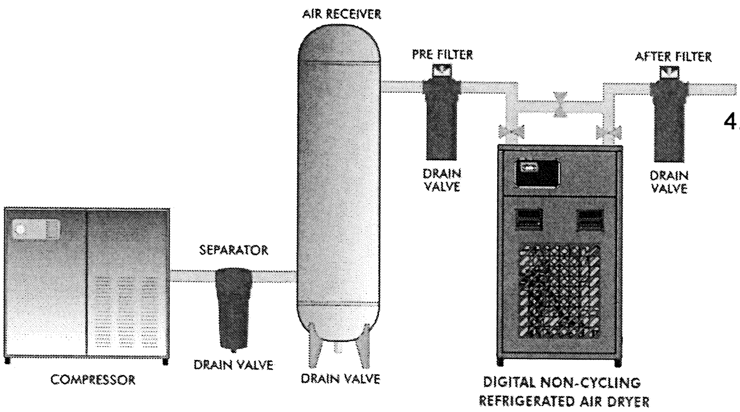
Microprocessor can be remotely controlled and any alarm contacts can be connected to any external devices.

2.5 PRINCIPLE OF OPERATION

The refrigerant circuit can be divided in 3 parts:

1. Low pressure section with an evaporator (heat exchanger)
2. High pressure section including: Condenser, liquid receiver, (if installed) and the filter dryer.
3. Control circuit including: Compressor, metering device, hot gas by-pass valve (if installed), fan pressure switch,

Figure 2-1: Typical Installation Diagram



safety high/ low pressure switch. Water cooled dryers are equipped with a water flow control valve Refrigeration Heat Exchanger

The compressor compresses gaseous refrigerant. The hot refrigerant gas condenses in the condenser and once liquefied it passes through the liquid receiver (optional)
The liquid refrigerant leaves the liquid receiver and is injected into the evaporator (heat exchanger) by a metering device (expansion valve).

This metering device is protected by a filter dryer, that retains particles and humidity that could be in the circuit

The injected liquid evaporates (changes states from liquid to gas) by absorbing heat from the compressed air. The gaseous refrigerant is sucked into the compressor and the cycle carries on.

In order to keep the evaporation pressure steady, and maintain the refrigerant temperature in the heat exchanger, a hot gas by-pass valve injects hot gaseous refrigerant into the low pressure refrigerant circuit.

2.6 COMPRESSED AIR HEAT EXCHANGER

1. The saturated, hot compressed air flows into the air/air heat exchanger where it is pre cooled by the out going chilled air.
2. The compressed air then enters the evaporator where it is cooled to the desired dew point.
3. The cooled compressed air passes through the air/water separator where the liquid condensate is removed by the centrifugal separation and drained away by the automatic trap. The outgoing, chilled air is then warmed up in the air/air heat exchanger by the hot incoming air. This energy efficient air-to-air heat exchanger significantly reduces the load on the compressor.
4. As long as the compressed air temperature does not drop below dew point, there will be no further condensation in the compressed air piping.

SULLAIR®

2.4 PRODUCT FEATURES

2.4.1 REFRIGERANT COMPRESSOR

The refrigerant compressor is fully hermetically sealed and requires no maintenance.

2.4.2 CONDENSER

1. Air cooled models: The refrigerant con- denser is equipped with fans that are cycled on and off to maintain a minimum high side pressure.
2. Water-cooled dryers Utilize a water control valve piloted by a sensing bulb that reacts to the high side refrigerant pressure controlling the water flow Maintain a minimum high side pressure.

2.4.3 REFRIGERANT CIRCUIT PROTECTION

1. **High / Low pressure safety switch**:

Refrigerant circuits are protected against excessive pressure by a safety switch that stops the compressor in cases of high or low refrigerant pressure. If this safety switch has tripped out on high refrigerant pressure, it has to be manually reset before the dryer can be restarted. A low pressure trip will automatically reset.

2.4.4 FILTER DRIER

Total water removal in the refrigerant circuit is achieved by a filter drier who also traps any solid particles that may have migrated into the circuit during assembly.

2.4.5 CRANKCASE HEATER

3-Phase dryers are equipped with an electric crankcase heater. The heater provides preliminary pre-heating of the refrigerant compressor to evaporate liquid refrigerant possibly condensed in the crankcase. This will prevent liquid shock that can damage the compressor.

⚠ WARNING

Never run the dryer before heating the crankcase. Allow 2 hours the heater to heat the compressor of the dryers after a long shut-down period.

2.4.6 REFRIGERANT CIRCUIT REGULATION IN (RF-A-10 – RF-A-7300)

1. The liquid refrigerant is injected into the evaporator through a metering device trying to maintain the refrigerant in the evaporator at a constant pressure
2. The evaporating pressure is kept constant by a controlled injection of hot gas from the high-pressure side into the low-pressure section of the circuit through a hot gas by- pass valve. This constant pressure corresponds to a stable evaporating temperature adjusted as close to 32°F / 0°C as possible.
3. The mixture of hot gas from the by-pass valve and cold gas from the evaporator is called superheat and is adjusted at 50 ± 5°F / 10± 5°C.

2.4.7 CONDENSATE DRAIN — TRAP ASSEMBLY

Dismantling the drain is easy because it can be isolated from the air circuit under pressure with a ball valve. Always isolate the drain before disassembly. MEMBRANEe disc may need to be replaced time to time due to contamination in the air circuit. Contact Sullair or its distributer if there is any drain blockage issues.

2.4.8 HEAT EXCHANGER, MODULAR DESIGN

The dryers are equipped with compact, modular design heat exchangers. This assembly has been specially designed to dry compressed air and consists of:

1. An air/air heat exchanger which pre-cools the entering hot air with the exiting chilled air.
2. An evaporator which is an air/refrigerant heat exchanger which cools down the com- pressed air.
3. An integral separator that separates the moisture from the air stream at the coldest point. Maximum condensation of moisture occurs at the coldest point and it is at that exact junction that the moisture is removed and drained.

2.4.9 DEW POINT INDICATOR

The Dew point indicator is a standard on all dryer models and is located in the control panel and provides an idea for the pressure dew point (PDP)

2.4.10 HIGH TEMPERATURE SWITCH

Located inside the dryer, this high temperature switch stops the dryers if the compressor suction line temperature is above 113°F / 45°C.

2.4.11 MICROPROCESSOR DEVICE

This device is supplied on all models.

RF-A SERIES DRYER

2.1 INTRODUCTION

The Sullair Non-cycling Refrigerated air dryers are designed for constant full load operation. They use specially designed heat exchangers with integrated components to provide consistent dewpoint and trouble-free service for years to come.

The dryers remove the moisture, from the compressed air stream. By using basic refrigeration, the hot saturated air is cooled in the highly efficient heat-exchangers and the moisture is condensed and removed. Additional filtration eliminates other contaminants and particles.

Water (moisture) is one of the greatest enemies of air tools and piping. Left untreated in the compressed air stream, this moisture will deteriorate the tools, equipment and piping with corrosion, pipe scaling, freezing and a host of other problems that will diminish the life of your entire compressed air system. With a Sullair refrigerant dryer, you can be assured of clean, dry air with consistent dewpoint between 36 and 40°F / 2 and 4°C. The unique three-in-one heat exchanger combination.

- **Air to Air** (Pre-cooler/re-heater Heat Exchanger)
- **Air to Refrigerant** (High efficiency Heat Exchanger)
- **Integrated Separator** (located at the coldest point to maximize moisture extraction)

heats up the exiting air, thereby eliminating the chances of freezing even in relatively cooler ambient conditions. Warm dry air will not affect the piping nor the tools and the result is extended tool life. In addition, applications that require clean dry air can rely upon Sullair's non-cycling refrigerated air dryers to provide dry air at full load continuously.

2.2 ENVIRONMENTAL PROTECTION

1. US/EU laws protect the environment against refrigerant being released into the atmosphere.
2. An annual leak control test at less than 5.0 gr/year should be performed by a qualified engineer if the refrigerant dryer contains more than 4.4 lbs / 2 kg of refrigerant. This control test has to be done twice a year if the dryer contains more than 66 lbs / 30 kg.
3. Prior to dryer disposal, the refrigerant must be properly recovered by a qualified engineer.

2.3 PRODUCT DESCRIPTION

This refrigerated compressed air dryer has been designed to remove water vapor from industrial compressed air that is free of any aggressive contaminants like ammonia, gaseous acid, dust, rust, liquid condensate, any other chemical or mineral substances capable of attacking or clogging the heat exchanger(s).

The optional, water-cooled condenser is not designed for use with seawater or water containing aggressive contaminants.

Please contact your factory representative for further questions on this issue. This dryer has been designed for indoor operation only.

The minimum and maximum values stated must be observed, as well as the safety precautions described in this manual.

NOTES

5. The minimum and maximum values listed on the data label must be strictly followed. If your application does not fall within the parameters indicated on the data label– **STOP**. Contact your Sullair representative/ distributor and clarify the issue before proceeding or operating the dryer.

6. All safety procedures and practices mentioned in this manual must be observed, as well as all of the federal, state and local safety precautions.

7. If any statement in this manual does not comply with the local legislation, the strongest standard is to be applied.

1.3 ISSUES TO AVOID

1. Any structural changes made to the dryer without the advise of Sullair or Sullair representatives will void all warranty.
2. Compressed air from this dryer does not meet the OSHA standards for breathing and hence it should not be used for breathing purposes. None of the protective alarms, equipment or devices should be tampered with. These protective items have been designed and installed on this dryer for your safety.
3. Do not operate the dryer at pressures temperatures of flow other than the ones mentioned on the data label.

SAFETY

The dryer has been designed and constructed in accordance with the generally recognized rules pertaining to refrigeration technology as well as industrial safety and accident prevention regulations.

The equipment design, development, production, assembly and customer service fall under the Sullair quality control system.

The dryer is state of the art. There are, however, hazards to the body, equipment and life accompanying this type of product if it is not operated for the purpose which it is intended by trained and specialized personnel.

The equipment supplied is intended exclusively for drying compressed air. Any other use or one exceeding this is considered unauthorized. Sullair cannot be held liable for damages resulting from incorrect or unauthorized use of the equipment. Any such risk is carried solely by the end user.

Authorized use means complete compliance with all of the conditions of operation, servicing and maintenance prescribed by Sullair in this Instruction and Operation Manual.

The dryer is only to be operated, serviced and repaired by trained personnel who are familiar with this type of equipment and understand fully its operation and any potential dangers.

1.1 SAFETY INFORMATION

The end user and operator must observe all National, State, and Local industrial and safety regulations dealing with the operation of pressure vessels under compressed air service. Also all "end user" safety rules for the same type of service must be adhered to. The following points list some of the important factors dealing with this type of equipment.

- Never make any constructional changes to the equipment.
- Use only original spare parts and accessories
- Never weld on any pressure vessel or modify it in any way.
- All maintenance on "pressure parts" must be carried out with the equipment shutdown, depressurized and locked out. Any in plant procedures or work permits regarding pressure vessels are to be adhered to.
- Do not operate the equipment with the control panel door open, the electrical system energized and live parts exposed.
- Disconnect the dryer from the electrical supply when any electrical work is performed. Lock out the safety disconnect and obtain any required work permits.

1.2 SAFETY INSTRUCTIONS

SAFETY REGULATIONS:

1. When operating the air dryer, the operator must adopt safe working methods and observe all local safety instructions and relevant regulations.
2. Prior to installation, the dryer and the compressed air system are to be depressurized and disconnected from the electrical main supply.
3. The user is responsible for safe operating conditions. Parts and accessories must be replaced if inspection shows that it is unsafe to operate the dryer.
4. Installation, operation, maintenance and repair are only to be performed by authorized, trained and skilled technicians.

CONTROLLER

MAINTENANCE

TECHNICAL DRAWINGS

SULLAIR®

TABLE OF CONTENTS

AIR CARE SEMINAR TRAINING

Sullair Air Care Seminars are courses that provide hands-on instruction for the proper operation, maintenance, and servicing of Sullair products. Individual seminars on Stationary compressors and compressor electrical systems are offered at regular intervals throughout the year at Sullair's training facility located at Michigan City, Indiana.

Instruction includes training on the function and installation of Sullair service parts, troubleshooting common faults and malfunctions, and actual equipment operation. These seminars are recommended for maintenance, contractor maintenance, and service personnel.

For detailed course outlines, schedule, and cost information contact:

SULLAIR TRAINING DEPARTMENT

1-888-SULLAIR or
219-879-5451 (ext. 5623)
www.sullair.com
training@sullair.com

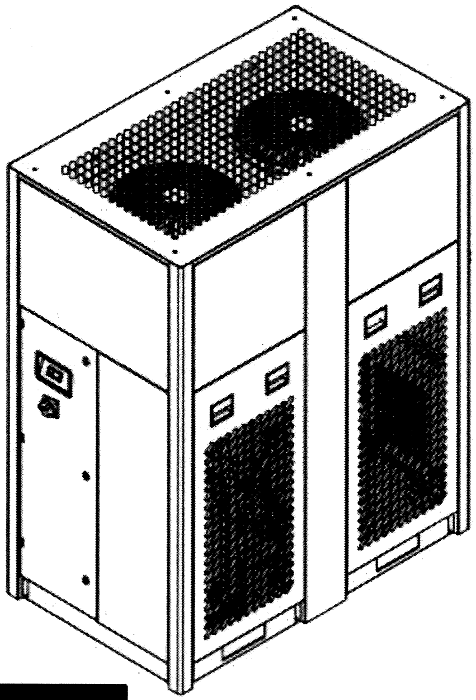
- Or Write -

Sullair

3700 E. Michigan Blvd.
Michigan City, IN 46360
Attn: Service Training Department.

USER MANUAL

REFRIGERATED DRYERS
RF-A—DIGITAL CYCLING



SAFETY WARNING

Users are required to read the entire User Manual before handling or using the product.

WARRANTY NOTICE

Failure to follow the instructions and procedures in this manual, or misuse of this equipment, will void its warranty.

The information in this manual is current as of its publication date and applies to Dryers manufactured after October 1, 2017 as indicated by

SERIAL NUMBER:
(RF-A) 1017MA00001

Publication Date: 10/01/2017

02250201-297 R01